

Personnel Zone Access and Alarm Response Procedures

Document: SOP-004 | Rev: 1.0 | Date: 2025-12-01 | Author: Ing. F. Ferrari — Safety & Compliance

1. Role-Zone Authorisation Matrix

Role	Prod. Area	Server Room	Gen. Warehouse	Main Entrance	Chem. Storage	R&D Lab	Cafeteria
operator	YES	NO	YES	YES	NO	NO	YES
technician	YES	YES	YES	YES	YES	YES	YES
supervisor	YES	YES	YES	YES	YES	YES	YES
manager	YES	YES	YES	YES	YES	YES	YES
security	YES	YES	YES	YES	YES	YES	YES
visitor	YES	NO	NO	YES	NO	NO	YES

2. Maximum Occupancy per Zone

Zone	Max Persons	Enforcement
Production Area	12	WARNING above 10; CRITICAL above 12
Server Room	2	CRITICAL above 2; access badge required
General Warehouse	4	WARNING above 3
Main Entrance	8	Monitoring only
Chemical Storage	4	CRITICAL above 4; hazmat training required — see RULE-CHM01-03
R&D Lab	6	WARNING above 5; precision experiments may be disturbed
Cafeteria	20	Monitoring only — occupant comfort

3. Maximum Acknowledgment Times by Role and Severity

Role	Max Ack — WARNING (min)	Max Ack — CRITICAL (min)
operator	15	3
technician	10	2
supervisor	5	1
manager	15	5
security	20	5

4. Unauthorised Access Rules

RULE-ACCESS-01: Any entry by an unauthorised role MUST be logged as `OccupancyEvent::UnauthorisedAccess` with `person_id`, `zone`, and `timestamp`.

RULE-ACCESS-02: Security personnel MUST be notified within 2 minutes of detection.

RULE-ACCESS-03: The KG node MUST include authorized=NO and a reference to this rule.

RULE-ACCESS-04: Chemical Storage has zero-tolerance policy. Any UNAUTHORIZED_ACCESS triggers immediate escalation to the Safety Officer regardless of role.